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Pseudomonas fluorescens and *Listeria monocytogenes* Planktonic Cells and Biofilms Are Inhibited by Surfactin from *Bacillus velezensis* H2O-1

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Abstract: Biofilms are highly important to be controlled in food industries for two major reasons: (i) pathogenic microorganisms can impact public health causing foodborne illness outbreaks, and (ii) food-spoilage microorganisms can cause economic impacts due to the loss of organoleptic quality. *Listeria monocytogenes* and *Pseudomonas fluorescens* are ubiquitous and highly representative of both problems. The presence of these bacteria in biofilms must be controlled, and new strategies need to be implemented. Among those strategies, the use of biosurfactants is promising. The present work studied the application of a surfactin produced by *Bacillus velezensis* H2O-1 to inhibit corrosion, planktonic growth, microbial adhesion, and biofilm formation by two strains of *L. monocytogenes* and one strain of *P. fluorescens*. For that purpose, scanning electron microscopy, epifluorescence microscopy, and the determination of the physicochemical characteristics of different surfaces, microorganisms and biofilms were performed. Biofilm reduction on conditioned surfaces reached up to 75%. When the surfactin was added to the media, the planktonic inhibition values reached 87%, and biofilms were inhibited by up to 100%. The analyzed images suggest that this molecule has great potential to postpone steel corrosion. The results demonstrated the great potential of this biomolecule in the food industry against both microorganisms, thus enhancing food safety and shelf-life.

Keywords: surfactin; biosurfactant; foodborne pathogen; *Listeria monocytogenes*; *Pseudomonas fluorescens*; antiadhesion; antibiofilm; surface modification

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1. Introduction

The microorganisms *Pseudomonas fluorescens* and *Listeria monocytogenes* are important for control in the food industry because they are usually associated with food spoilage and consumer health, respectively [1,2]. These bacteria are ubiquitous and difficult to eradicate from food plants once they are installed because of their ability to grow under a wide range of conditions and their ability to form biofilms [2,3].

The presence and persistence of pathogenic microorganisms in food plants may lead to food contamination. Depending on the species and strains and how the consumer will prepare the food prior to consumption, an outbreak of food disease can be widespread;

therefore, it is important to detect their presence and develop methodologies to prevent their establishment on food contact surfaces [4–10].

According to the World Health Organization [11], *Listeria monocytogenes* is a relevant pathogen associated with listeriosis disease. This illness has a high mortality rate (up to 30%) and is usually associated with ready-to-eat products, leading to the necessity of food regulations that do not tolerate its simple presence in such types of products.

When a food-spoilage microorganism is present and well-established in food plants, the impact will not be on public health but rather on the shelf-life of the food products, leading to economic loss due to the poor organoleptic quality of the products. Usually, food-spoilage microorganisms are ubiquitous in the environment and are naturally found in raw products, hindering their control in the food-industry [2,12–15].

Pseudomonas fluorescens is one of the most common microorganisms that causes food spoilage and forms biofilms on many surfaces in the food industry. It produces lipolytic and proteolytic enzymes that are heat resistant, and it decreases the quality and shelf-life of products [16].

Biofilms are a form of resistance of microorganisms to environmental conditions. First, planktonic cells attach to surfaces; then, they become irreversibly adhered and begin to secrete exopolymeric substances (EPSs), forming a matrix that is responsible for maintaining nutrient exchange within the biofilm [17]. Therefore, cells become protected from outside environmental conditions, and when they mature, plugs and cells are released from the matrix and can colonize other sites [17,18].

Owing to the difficulty of eliminating biofilms from food plants, it is important to develop strategies associated with current techniques to overcome the presence and development of such microorganisms in food contact surfaces [19,20]. One strategy that can be used is to condition surfaces for antiadhesion and antibiofilm effects, and biosurfactants have been demonstrated to have great potential for use in such strategies [21,22].

Biosurfactants (BSs) are amphipathic molecules produced by microorganisms; these molecules have antimicrobial activity and can change the interfacial properties of surfaces, leading to microbial control [23,24]. Among those substances is surfactin, which is a powerful lipopeptide surfactant composed of seven amino acids linked to straight or branched fatty acid chains that are 13 to 16 carbons in length [25], and it has great potential for use in the food industry to avoid food contamination [25–27].

This substance is produced by a wide variety of *Bacillus* species, mainly *Bacillus subtilis*, and little is known about its production and potential applications than other *Bacillus* species. *Bacillus velezensis* is a surfactin producer; this microorganism (strain H2O-1) was previously isolated by our group from connate water from a Brazilian reservoir [28]; its surfactin was characterized, and its production was optimized [29,30]. It is stable over a wide range of pH values, temperatures, and salinities and was considered safe for environmental applications in acute toxicity tests [29–31].

Therefore, the aim of this study was to evaluate surfactin produced by *B. velezensis* H2O-1 as a surface conditioning agent to prevent the adhesion and biofilm formation of *Pseudomonas fluorescens* ATCC 13525 and *Listeria monocytogenes* ATCC 7644 and ATCC 19112 on polystyrene and steel surfaces. Additionally, the antimicrobial activity on planktonic cells and the anticorrosion behavior were assessed.

2. Materials and Methods

2.1. Microorganisms

The surfactin-producing strain *Bacillus velezensis* H2O-1 (Laboratório de Genética Microbiana, Rio de Janeiro, Brazil), the strains *L. monocytogenes* ATCC 7644 and ATCC 19112 (Instituto Nacional de Controle de Qualidade em Saúde—INCQS, Rio de Janeiro, Brazil), and *P. fluorescens* ATCC 13525 (INCQS, Rio de Janeiro, Brazil) were stored in LB broth (10 g/L tryptone, 5 g/L yeast extract, and 5 g/L NaCl) supplemented with 20% (v/v) glycerol at −80 °C until use.

2.2. Surfactin Production and Characterization

Surfactin from *B. velezensis* H2O-1 [28] was produced, purified, and characterized according to the methodology described by Guimarães et al. [31]. After production in mineral salt medium containing (w/v) NaCl 1.0, Na_2HPO_4 0.5, KH_2PO_4 0.2, $MgSO_4$ 0.02, and $(NH_4)_2SO_4$ 0.2, with glucose as the only carbon source (1.0% w/v), for 72 h at 30 °C and 170 rpm, the cells were removed via centrifugation (12,500× g, 4 °C for 20 min), and the supernatants were sterilized, filtered (0.22 µm), and purified for further analysis. High-performance liquid chromatography (HPLC) (Agilent Technologies 1200, Santa Clara, CA, USA) coupled with a UV detector at 210 nm and a C18 column (150 × 4.6 mm, 5-µm particle size; ZORBAX, Agilent, Santa Clara, CA, USA) was used to quantify surfactin according to Guimarães et al. [29].

2.3. Surface Conditioning

The surface conditioning was performed according to Araujo et al. [32], where the cleaned surfaces [33] of microtiter plate wells and square coupons (20 × 20 mm) of polystyrene, stainless steel (AISI 304 and AISI 430), galvanized steel, and carbon steel were preconditioned with aqueous solutions of the surfactant at different concentrations (5000, 2500, 1000, and 500 ppm). Water was used as a negative control, and sodium dodecyl sulfate (SDS) was used as a positive control. After 24 h, the surfaces were washed with sterile water and dried at room temperature.

2.4. Physicochemical Properties

Surfactin interfacial properties were analyzed via the pendant drop technique (DSA 100S Goniometer, Model: OF 3210, Krüss, Hamburg, Germany) according to Song and Springer [34] for the determination of surface tension (ST), interfacial tension (IT, against n-hexadecane), and critical micelle concentration (CMC), and the results are expressed as the mean values of at least ten pendant drops at 23 °C and 55% relative humidity. To enable the determination of the CMC, surfactin serial dilutions in distilled water were performed according to Sheppard and Mulligan [35] prior to the surface tension measurements. The emulsification index (E_{24}) of surfactin (n-hexadecane) was evaluated according to Cooper [36], and the height of the emulsified layer was monitored for 30 days to check for emulsion stability.

To assess the different surface interfacial properties (surface free energy and surface hydrophobicity), the sessile drop technique (Krüss DSA 100S goniometer, model OF 3210, Krüss, Hamburg, Germany) was used [32]. The contact angle between the targeted surfaces and water, formamide and ethylene glycol was measured according to Van Oss [37] and Van Oss et al. [38], and the results are expressed as the mean value of at least ten drops (10 µL) at 23 °C and 55% relative humidity.

2.5. Microscopy Analysis: Epifluorescence and Scanning Electron Microscopy (SEM)

The sample surfaces were examined by scanning electron microscopy (SEM) to evaluate biofilms and the extent of corrosion of those metal alloys. For the corrosion analysis, biofilms were removed from the samples, and they were cleaned according to the methods of Parizzi et al. [39] prior to metallization (polystyrene) and examined via high-vacuum scanning electron microscopy (FEI Company Quanta 200, Dawson Creek Drive, Hillsboro, OR, USA), where the images were obtained at a magnification of 1500 × 20 kV. For biofilm analysis, the samples were washed in sodium cacodylate buffer (0.1 M), dehydrated in an ethanol gradient, and dried at the CO_2 critical point. The samples were subsequently metallized (~5 nm) prior to examination via the same high-vacuum scanning electron microscope at a magnification of 5000 × 20 kV. For epifluorescence microscopy, the bacterial viability kit L7012 LIVE/DEAD® *Baclight*™ (Molecular Probes, Inc., Eugene, OR, USA) was used, following the instructions of the supplier, and the samples were observed under a microscope with an epifluorescence system (Axioplan 2 microscope, Carl Zeiss Industrielle Messtechnik GmbH, Oberkochen, Germany) using filter set 09 and a digital camera (Color View XS, AnalySIS GmBH, Münster, Germany).

2.6. Biofilm and Planktonic Growth Analyses

To develop and quantify biofilms, the procedure was carried out as described by Araujo et al. [32], where bacterial suspensions from *L. monocytogenes* and *P. fluorescens* were standardized by optical density (610 nm, ANALYTIK JENA SPECORD 210, Langewiesen, Ilmenau, Thüringen, Germany) to achieve 10^9 CFU/mL. Then, 10% aliquots of these suspensions were inoculated in nutrient broth and incubated at 35 °C and 30 °C (*L. monocytogenes* and *P. fluorescens*, respectively), and the kinetics of planktonic growth, adhesion, and biofilm formation on the surfaces were monitored.

To quantify biofilms, after culture removal, the surfaces were washed to remove nonadherent cells and fixed with methanol prior to staining with 1% crystal violet for 20 min, according to Stepanovic et al. [40]. The stain was resuspended in 33% glacial acetic acid, and the optical density was verified at 570 nm.

To correlate the biofilms' optical density readings with their weight, a dry mass assay was performed as described by Cerca et al. [41], where biofilms from known surface areas (cm^2) were scraped, resuspended, and filtered through preweighed cellulose acetate membranes (0.22 µm). After drying at 80 °C until a constant weight was reached, the dry weight (gbiofilm) was subsequently converted into gbiofilm/cm^2. The planktonic growth in the presence and absence of surfactin (added to the media or as a surface condition agent) was directly quantified by optical density (610 nm).

The biofilm composition (exopolysaccharides and protein) was determined after separating the matrix from the biofilms via ultrasonication (30 W, 30 s on ice) followed by 2 min of stirring in a vortex mixer, centrifugation for 10 min at 4 °C at 3000× *g*, and filtering through 0.22 µm membranes to remove the cells. The matrix content is the total dry weight of the biofilm over the dry weight of filtered cells ($mg_{matrix}/g_{biofilm}$). Exopolysaccharides were determined according to Dubois et al. [42], using standard curve glucose, and proteins were quantified as described by Bradford [43] with bovine serum albumin (BSA) as the standard curve. Both components were subsequently normalized per area of biofilm (µg/cm^2).

2.7. Statistics

The results were analyzed via ANOVA to check for significant differences between the treated and non-treated groups, and the means were subsequently compared via Tukey's test with 5% of probability via Statistica 7 software (STATSOFT Inc., Hamburg, Germany).

3. Results

3.1. Physicochemical Characteristics of Surfactants

Bacillus velezensis H2O-1 produced 802.6 ± 3.38 mg/L surfactin. This product achieved an emulsification index of 47.0 ± 0.3%, and the emulsion did not present statistically significant differences during the studied period of 30 days, whereas SDS reached 70.6 ± 0.9%, but the emulsion decreased by 11% during the same period. The surfactin sample presented the lowest surface tension (SFT) and interfacial tension (IFT) values, with values of 24.8 ± 0.2 mN/m and 0.8 ± 0.2 mN/m, respectively. The commercial surfactant SDS presented higher SFT and IFT values: 35.5 ± 0.3 mN/m and 7.5 ± 0.1, respectively. The critical micellar concentration (CMC) was also lower for surfactin than for SDS: 38.7 ± 0.3 and 891.1 ± 11.0 mg/L, respectively.

3.2. Physicochemical Characteristics of Microorganisms and Surfaces

All the tested microorganisms were strong biofilm formers, with *P. fluorescens* biofilms presenting the greatest adhesion and biofilm development on the tested surfaces. Figure S1 (in the Supplementary Materials) shows the mature biofilms of the three tested microorganisms on polystyrene and stainless steel AISI 304. Among the tested microorganisms, the *L. monocytogenes* strain ATCC 7644 had the lowest biofilm formation ability.

Table 1 and Figure S2 (in the Supplementary Materials) present the physicochemical properties of the microorganisms and surfaces. The most hydrophobic microorganism was *L. monocytogenes* ATCC 7644, and the least hydrophobic microorganism was *P. fluorescens* ATCC 13525. The surface treatments involving conditioning with SDS and surfactin changed the physicochemical parameters, indicating that the adsorption of those molecules to the surfaces, therefore, was able to change the behavior of adhesion and biofilm formation and protect the surfaces from corrosion. The corrosion of the stainless steel AISI 304, stainless steel AISI 430, galvanized steel, and carbon steel was lower with surfactin treatments than with the unconditioned surfaces and the SDS-conditioned surfaces, as shown in Figures S3–S6 (in the Supplementary Materials).

Table 1. Physicochemical properties of the surfaces of polystyrene, stainless steel (SS) AISI 304, stainless steel (SS) AISI 430, galvanized steel (GS), and carbon steel (CS): surface total free energy (γ^{TOT}); Lifshitz–van der Waals component (γ^{LW}); Lewis acid–base property (γ^{AB}); electron donor component (γ^{-}); electron acceptor component (γ^{+}); and surface hydrophobicity (ΔG_{iWi}). Figure S2 reproduces the physicochemical data concerning the microorganisms (*L. monocytogenes* ATCC 19112 and ATCC 7644 and *P. fluorescens* ATCC 13525) and the polystyrene and stainless steel AISI 304 unconditioned surfaces from our previous study [32].

Microorganism/Surface	γ^{LW} (mJ/m^2)	γ^{-} (mJ/m^2)	γ^{+} (mJ/m^2)	γ^{AB} (mJ/m^2)	γ^{TOT} (mJ/m^2)	ΔG_{iWi} (mJ/m^2)
Polystyrene × Surfactin	40.7 ± 0.1	0.0 ± 0.1	6.0 ± 1.6	0.1 ± 0.5	40.8 ± 0.6	−89.7 ± 4.7
Polystyrene × SDS	49.1 ± 0.1	0.5 ± 0.1	22.3 ± 0.1	−6.8 ± 0.1	42.3 ± 0.1	−27.0 ± 3.6
SS AISI 304 × Surfactin	69.0 ± 1.2	1.7 ± 0.4	57.2 ± 0.5	−19.5 ± 1.6	49.5 ± 1.2	37.7 ± 5.1
SS AISI 304 × SDS	35.2 ± 1.3	0.3 ± 0.2	0.9 ± 0.6	0.9 ± 0.7	36.1 ± 1.3	−115.8 ± 5.3
SS AISI 430 (unconditioned)	29.0 ± 0.2	2.6 ± 2.2	1.6 ± 1.3	−4.1 ± 1.5	25.0 ± 0.8	−119.1 ± 4.7
SS AISI 430 × Surfactin	62.1 ± 0.4	0.4 ± 0.2	41.0 ± 2.1	−7.9 ± 0.9	54.2 ± 2.4	20.6 ± 2.7
SS AISI 430 × SDS	34.7 ± 0.6	0.1 ± 0.2	4.2 ± 1.9	−1.1 ± 0.7	33.6 ± 0.4	−103.2 ± 4.5
GS (unconditioned)	35.1 ± 0.2	0.7 ± 0.5	4.9 ± 1.6	−3.6 ± 0.9	31.5 ± 1.0	−97.9 ± 4.5
GS × Surfactin	52.6 ± 0.5	9.8 ± 4.7	105.9 ± 6.8	−64.5 ± 2.4	−11.9 ± 2.9	40.1 ± 4.9
GS × SDS	28.9 ± 0.4	0.9 ± 0.9	31.1 ± 8.7	10.7 ± 6.6	39.6 ± 7.0	−9.3 ± 2.5
CS (unconditioned)	75.6 ± 0.4	2.8 ± 2.5	40.7 ± 14.5	−21.4 ± 5.5	54.2 ± 5.5	17.7 ± 2.3
CS × Surfactin	69.6 ± 0.4	1.8 ± 0.9	49.7 ± 2.9	−18.8 ± 1.4	50.8 ± 3.2	28.3 ± 3.1
CS × SDS	71.4 ± 0.3	2.2 ± 1.5	31.0 ± 6.1	−16.5 ± 4.1	54.9 ± 4.4	−2.9 ± 3.3

Table 2 shows that the percentages of cells in mature biofilms were approximately 71.6%, 85.4%, and 83.3% and the contents of sugar and protein within the biofilms were

20.5%, 9.1%, and 14.1% for *P. fluorescens* ATCC 13525, *L. monocytogenes* ATCC 19112, and *L. monocytogenes* ATCC 7644, respectively.

Table 2. Biofilm composition (sugars, proteins, matrix, and cells) per cm^2 for each strain of microorganism: *L. monocytogenes* ATCC 7644, *L. monocytogenes* ATCC 19112, and *P. fluorescens* ATCC 13525.

Microorganism	Matrix Sugar Content (μg/cm^2)	Matrix Protein Content (μg/cm^2)	Matrix Content (μg/cm^2)	Cells (μg/cm^2)
ATCC 7644	287 ± 1.9	384 ± 16.8	796 ± 199 *	3969 ± 189 *
ATCC 19112	182 ± 16.4	193 ± 19.7	605 ± 134 *	3538 ± 294 *
ATCC 13525	672 ± 3.7	836 ± 70.3	2091 ± 198	5272 ± 179

* Data do not differ statistically.

The results of interfacial energy adhesion (system free energy) between the surfaces (conditioned or not) and the microorganisms are shown in Table 3.

Table 3. Free interfacial energy of adhesion between the surfaces (conditioned and unconditioned) and microorganisms (*L. monocytogenes* ATCC 19112, *L. monocytogenes* ATCC 7644, and *P. fluorescens* ATCC 13525).

Treatment	*L. monocytogenes* ATCC 19112 ΔF_{adh}	*L. monocytogenes* ATCC 7644 ΔF_{adh}	*P. fluorescens* ATCC 13525 ΔF_{adh}
Polysterene	−15.2 ± 3.5	4.6 ± 1.4	−20.2 ± 1.3
Polysterene + Surfactin	−7.8 ± 5.4	17.8 ± 3.3	−14.3 ± 3.2
Polysterene + SDS	−27.1 ± 4.9	4.3 ± 2.8	−36.1 ± 2.6
Stainless Steel AISI 304	−2.9 ± 4.2	7.3 ± 2.1	2.9 ± 2.0
Stainless Steel AISI 304 + Surfactin	−10.8 ± 5.6	−9.6 ± 3.5	−64.3 ± 3.4
Stainless Steel AISI 304 + SDS	−6.1 ± 5.7	17.6 ± 3.6	2.3 ± 3.5
Stainless Steel AISI 430	−3.3 ± 5.4	6.3 ± 3.3	−0.5 ± 3.2
Stainless Steel AISI 430 + Surfactin	−3.6 ± 4.4	0.1 ± 2.3	−52.9 ± 2.2
Stainless Steel AISI 430 + SDS	6.2 ± 5.3	16.4 ± 3.2	−9.2 ± 3.1
Galvanized Steel	1.6 ± 5.3	10.8 ± 3.2	−10.6 ± 3.1
Galvanized Steel + Surfactin	−30.1 ± 5.5	−35.0 ± 3.4	−90.8 ± 3.3
Galvanized Steel + SDS	−5.2 ± 4.3	−0.9 ± 2.2	−44.4 ± 2.1
Carbon Steel	−12.0 ± 4.2	−9.5 ± 2.1	−52.1 ± 2.0
Carbon Steel + Surfactin	−10.3 ± 4.6	−8.4 ± 2.5	−59.1 ± 2.4
Carbon Steel + SDS	−9.1 ± 4.7	−5.2 ± 2.6	−44.1 ± 2.5

3.3. Antimicrobial, Antiadhesion, and Antibiofilm Properties

The surfactin- and SDS-conditioned surfaces of polystyrene and stainless steel AISI 304 presented antiadhesion, antibiofilm, and antimicrobial properties, as shown in Figure 1.

When surfactin was used as a conditioning agent on polystyrene surfaces, the inhibition of adhered cells (Figure 1A) reached approximately 55.0 ± 15.7%, 52.9 ± 20%, and 60.0 ± 7.3% for *L. monocytogenes* ATCC 7644, *L. monocytogenes* ATCC 19112, and *P. fluorescens* ATCC 13525, respectively. On the stainless-steel surfaces, this inhibition was 51.9 ± 7.9% and 74.6 ± 3.1% for *L. monocytogenes* ATCC 19112 and *P. fluorescens* ATCC 13525, respectively, whereas this surface treatment did not significantly inhibit the adhesion of the *L. monocytogenes* ATCC 7644 strain. After the biofilms matured (Figure 1B), the inhibition observed on polystyrene was approximately 36.7 ± 6.5%, 16.9 ± 13.6%, and 27.5 ± 8.5% for *L. monocytogenes* ATCC 7644, *L. monocytogenes* ATCC 19112, and *P. fluorescens* ATCC 13525, respectively, and that on stainless steel, AISI 304 reached 71.2 ± 8.4%, 62.9 ± 8.9%, and 40.3 ± 7.3% for *L. monocytogenes* ATCC 7644, *L. monocytogenes* ATCC 19112, and *P. fluorescens* ATCC 13525, respectively.

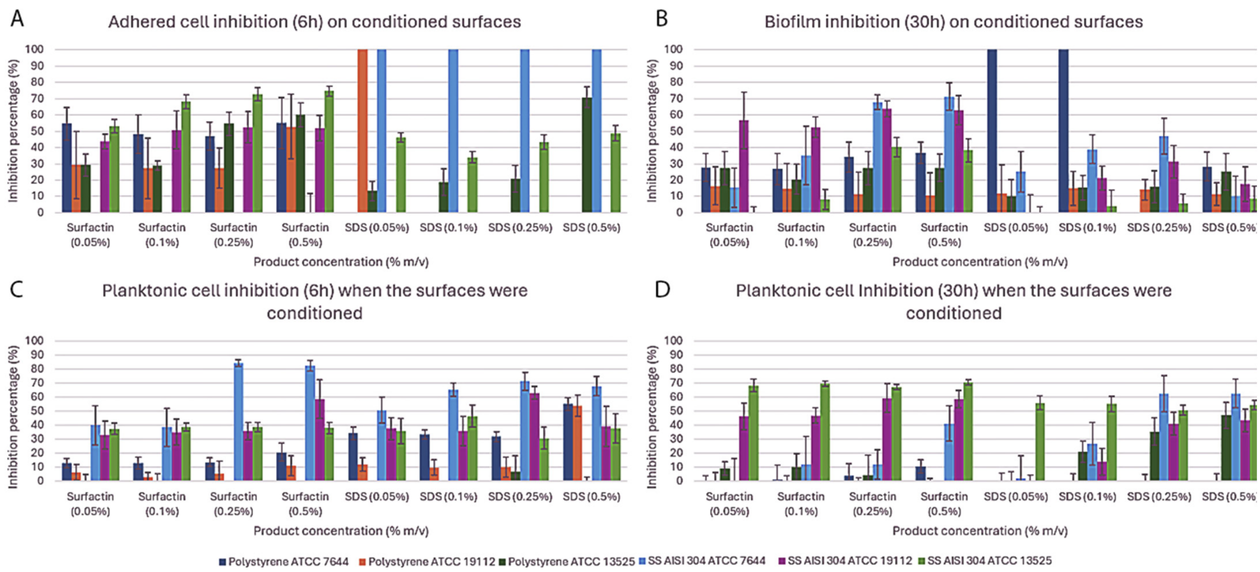


Figure 1. *L. monocytogenes* ATCC 7644 and ATCC 19112 and *P. fluorescens* ATCC 13525 percentage of adhered cell inhibition (**A**), biofilm inhibition (**B**), and planktonic cell inhibition (**C**,**D**) when surfactants (SDS and surfactin) were used as surface conditioning agents with concentrations ranging from 0.05 to 0.50% (m/v).

When SDS was used as a conditioning agent on polystyrene surfaces, adhered cell inhibition (Figure 1A) was not effective for the *L. monocytogenes* ATCC 7644 strain; however, with the other strain of *L. monocytogenes* (ATCC 19112), it was effective only when the lowest concentration was used, and with the *P. fluorescens* ATCC 13525 strain, the maximum adhesion inhibition was 70.9 ± 6.3%. On stainless steel surfaces, this treatment effectively inhibited the strain of *L. monocytogenes* ATCC 7644 but did not significantly inhibit adhesion to the strain of *L. monocytogenes* ATCC 19112, and the maximum inhibition for the strain of *P. fluorescens* ATCC 13525 was 48.8 ± 4.8%. After the biofilms matured (Figure 1B), the two lowest concentrations (0.05 and 0.1% m/v) used to condition the surfaces effectively inhibited 100 ± 13.5% of the biofilm development from both strains of *L. monocytogenes*, and the maximum inhibition obtained for the *P. fluorescens* ATCC 13525 strain was 25.2 ± 11.2% at the maximum tested concentration (0.5% m/v). On stainless steel AISI 304 surfaces, no significant inhibition was observed for the *P. fluorescens* ATCC 13525 strain, whereas it was able to decrease the biofilm development of *L. monocytogenes* ATCC 7644 and *L. monocytogenes* ATCC 19112 to 47.0 ± 11.2% and 31.3 ± 10.1%, respectively.

Epifluorescence microscopy with staining for biofilm viability confirmed adhesion and biofilm inhibition. A greater number of viable cells were present on the control surfaces of both polystyrene and AISI 304 stainless steel than on the surfaces treated with surfactin and with SDS, as shown in Figure 2 (*L. monocytogenes* ATCC 7644).

The conditioning of polystyrene surfaces with surfactin significantly interfered with the planktonic growth of *L. monocytogenes* ATCC 7644 and ATCC 19112, resulting in 20.8 ± 6.3% and 11.1 ± 8.7% inhibition at 6 h, respectively (Figure 1C) and 10.6 ±4.6% and 37.5 ± 2.7% inhibition at 30 h (Figure 1D), respectively. This same product inhibited up to 84.4 ± 2.3%, 58.8 ± 13.8%, and 38.7 ± 2.9% (ATCC 7644, ATCC 19112, and ATCC 13525, respectively) after 6 h (Figure 1C) and 41.1 ± 12.6%, 59.3 ± 10.1%, and 70.4 ± 2.0% (ATCC 7644, ATCC 19112, and ATCC 13525, respectively) after 30 h (Figure 1D) when the stainless steel AISI 304 surfaces were treated.

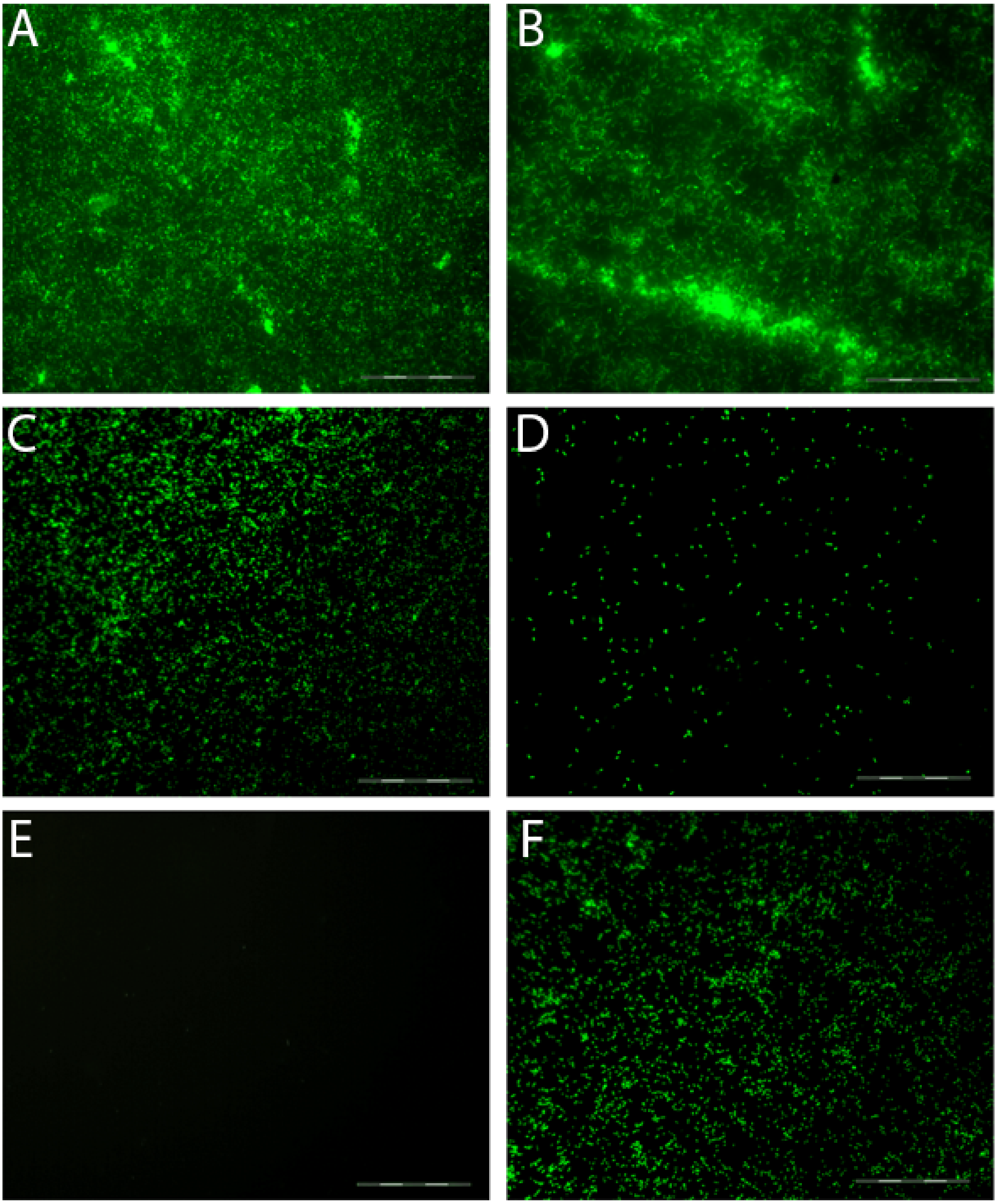
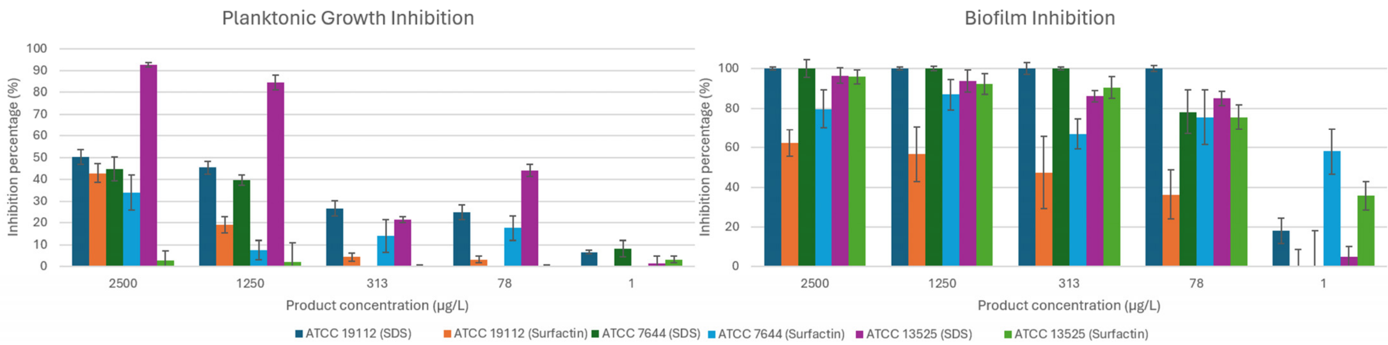


Figure 2. Epifluorescence microscopy of *L. monocytogenes* ATCC 7644 on polystyrene (**A**) and stainless steel (**B**) surfaces after 30 h of incubation, and their surfaces treated with surfactin (**C**,**D**) and SDS (**E**,**F**); the bars represent 200 µm. The viable cells are green, and dead cells are red.

Planktonic growth was also inhibited when the surfaces were conditioned with SDS, reaching maximum values of 55.2 ± 4.6% and 54.0 ± 7.5% (ATCC 7644 and ATCC 19112, respectively) after 6 h (Figure 1C) and 47.1 ± 9.2% (ATCC 13525) after 30 h (Figure 1D) with polystyrene-treated surfaces. With the stainless steel AISI 304-treated surfaces, planktonic growth was inhibited to 71.3 ± 6.5%, 63.3 ± 4.4%, and 46.4 ± 7.8% (ATCC 7644, ATCC 19112, and ATCC 13525, respectively) after 6 h (Figure 1C) and to 62.6 ± 10.3%, 43.4 ± 7.9%, and 56.0 ± 5.0%, respectively, after 30 h (Figure 1D).

When directly added to the culture media instead of conditioning the surfaces, both products (SDS and surfactin) were able to interfere with biofilm development and planktonic cell growth, as shown in Figure 3.



(**A**) (**B**)

Figure 3. *L. monocytogenes* ATCC 19112 and ATCC 7644 and *P. fluorescens* ATCC 13525 percentage of planktonic cells (**A**) and biofilm (**B**) inhibition when the surfactants (SDS and surfactin) were added to the culture media with concentrations varying from 1 to 2500 μg/L.

When surfactin was added to the culture media, planktonic growth (Figure 3A) of both strains of *Listeria monocytogenes* was inhibited to 42.9 ± 4.3% and 33.9 ± 8.1%, respectively, with 2.5 g/L surfactant, and biofilm growth (Figure 3B) was inhibited to 64.4 ± 6.6% and 86.8 ± 9.5% for strains ATCC 19112 and 7644, respectively. None of the tested concentrations were able to reduce the planktonic growth of *Pseudomonas fluorescens,* but its biofilm growth was inhibited from 35.7 ± 7.1 to 95.7 ± 3.4%, with concentrations varying from 0.001 to 2.5 g/L.

When SDS was added to the culture media, it effectively reduced the planktonic growth (Figure 3A) of both strains of *L. monocytogenes* and *P. fluorescens*, achieving maxima of 50.3 ± 3.3, 44.8 ± 5.3, and 92.5 ± 1.1% for ATCC 19112, ATCC 7644, and ATCC 13525, respectively. The biofilm formation of these strains (Figure 3B) was also inhibited by 100 ± 0.8%, 100 ± 4.5%, and 96.3 ± 3.9%, respectively.

4. Discussion

Biosurfactants usually present better interfacial properties (lower IFT, SFT, CMC, and emulsion stability) than commercial surfactants do and are more effective under a variety of environmental conditions [44,45]. As expected, those properties were more effective for the tested surfactin than for the SDS. Although its production and productivity still need to be enhanced, this molecule presents excellent potential for use at low concentrations, decreasing the need for high amounts for each application [45].

The adhesion and biofilm formation kinetics corroborated the physicochemical parameters (Table 1), which revealed that the first microbial adhesion phase, according to the Lifshitz–van der Waals component, occurred in *P. fluorescens* ATCC 13525 on carbon steel, followed by stainless steel AISI 304, polystyrene, galvanized steel, and stainless steel AISI 430, and the lowest interaction occurred with *L. monocytogenes* ATCC 7644 on polystyrene.

The surfaces with low free energy are hydrophobic, and at hydrophobic surfaces, hydrophobic interactions are considered the main forces; at hydrophilic surfaces, electrostatic interactions are the main forces [46,47]. As the microorganisms and surfaces are hydrophobic, the strong hydrophobic interactions explain the establishment of well-developed biofilms over the unconditioned tested surfaces.

Jemil et al. [48] tested the adhesion and biofilm inhibition of *Staphylococcus aureus*, *Salmonella typhimurium*, *Klebsiella pneumoniae*, *Candida albicans*, and *Bacillus cereus* by conditioning polystyrene surfaces with a lipopeptide and reported that biofilm inhibition is concentration-dependent, whereas the antiadhesive effect usually remains constant. They reported that the inhibition of biofilm formation was due to the ability to change the physicochemical properties of the surface, especially the electrostatic repulsion forces.

When the conditioned surfaces became more hydrophilic, the number of hydrophobic interactions with the cell surfaces decreased, leading to a decrease in adhesion/biofilm formation. The hydrophilic characteristics of microorganism surfaces are related to high amounts of polysaccharides and low amounts of hydrocarbon compounds [49]; therefore, if a strain can produce exopolysaccharides, it will also be less hydrophobic and will present a greater ability to form biofilms [32], as found in *P. fluorescens* ATCC 13525 (Table 2, Figure S1). Additionally, this same strain presented the greatest amount of matrix (28.4%), and among the matrix components—72.1% were due to the presence of sugar and protein, accounting for 20.5% of the total biofilm composition. *Pseudomonas* sp. are known to be lipopolysaccharides producers, and some of these compounds are important for adhesion and biofilm development over a variety of surfaces [50]; here, surfactin-preconditioned surfaces affected *P. fluorescens* ATCC 13525 adhesion more intensely than mature biofilms from the same strain (Figure 1A,B).

Cervantes-Huamán et al. [51] studied the matrix composition of *L. monocytogenes* CECT 5672, *S. typhimurium* CECT 4594, and *S. aureus* CECT 239 biofilms on stainless steel and verified that all tested strains presented a greater percentage of proteins (approximately 83%) than polysaccharides and eDNA did. Here, proteins were also the major components of the matrix, followed by sugars and nonidentified components.

The metallic surfaces became more hydrophilic when treated with surfactin (Table 1). When the biosurfactant adsorbs to a surface, it usually exposes the hydrophilic group to the bulk and the hydrophobic group to the surface, making the surface more hydrophilic and leading to a decrease in the interfacial tension. This film layer can also be responsible for surface protection against corrosion [52].

Adhesion may occur if it results in a decrease in the system free energy (if the value is less than zero, it will adhere to the surface) [53,54]. The treatments that lowered the free energy (Table 3) were less effective at inhibiting adhesion (6 h) (Figure 1A). Adhesion and biofilm development were significantly reduced with surfactin treatment (Figure 2C,D). These results corroborated the idea that the surfactin produced by this strain of *B. velezensis* H2O-1 also presents great potential for antiadhesion and antibiofilm formation, as other *Bacillus* sp. surfactin producers do [32].

Jardak et al. [55] studied the antibacterial, antiadhesive, and antibiofilm effects of a surfactin produced by *Bacillus* sp. 15F. With respect to antimicrobial properties, they reported the most potent activity against *S. epidermidis* S61, whereas with respect to *L. monocytogenes*, they reported values ranging from 0.5 to 1.0 mg/L as the minimal inhibitory concentration (MIC) and reported that this property is due to the membrane penetration of surfactin, leading to the dissolution of the membrane cells. The biofilm eradication effect reached 99.9% when 62.5 μg/L surfactin was added to preformed biofilms, which is attributed to the same mechanism used for planktonic growth inhibition, whereas the antiadhesion mechanism of surfactin against *S. epidermidis* S61 (90% with 62.5 μg/L) was attributed to the ability to change the surface hydrophobicity via adsorption.

Here, the antibiofilm inhibition mechanism when the surfaces were conditioned is probably related to an association between antimicrobial activity and changes in the physicochemical properties of the surfaces, leading to adhesion inhibition and subsequent biofilm inhibition. The desorption of surfactin from surfaces also affects planktonic growth, leading to planktonic cell inhibition when the products are used as conditioning agents (Figure 1C,D).

The antimicrobial effect of adding surfactin to the culture media (Figure 3A) was greater for Gram-positive microorganisms (*L. monocytogenes* ATCC 7644 and ATCC 19112) than for Gram-negative microorganisms (*P. fluorescens* ATCC 13525), probably due to differences in their cellular structure. Gram-positive bacteria are usually highly sensitive to anionic detergents, whereas Gram-negative bacteria are not [56].

Surfactin has been demonstrated to have antibacterial activity against Gram-positive bacteria, but the same is not true for Gram-negative bacteria [57,58]. Additionally, Zhang et al. [59] corroborated the idea that surfactants (quaternary ammonium compounds) are more effective against Gram-positive than Gram-negative microorganisms because external membrane layers act as an extra barrier in Gram-negative microorganisms. Therefore, it can be inferred that the mechanism of biofilm inhibition of *P. fluorescens* ATCC 13525 when surfactin was added to the bulk media is due to physicochemical properties rather than primarily antimicrobial activity while the combination of antimicrobial activity with physicochemical properties would be responsible for the biofilm inhibition of both strains of *L. monocytogenes*.

According to Boufioux et al. [60], surfactin more intensely affects membranes containing the short chains or fluid organization of phospholipids by interfering with their biological functions through their insertion in the lipidic bilayer, permeability modification by ionic channel formation or by carrying monocations and divalent cations and membrane solubilization via its detergent mechanism. Dai et al. [61] reported that the surfactin effect is dose-dependent and induces severe damage due to cell wall and/or cell membrane disintegration, leading to cell leakage until death. Besides the detergent-like model, surfactin can also inhibit protein synthesis in pathogenic bacteria, preventing cell reproduction and inhibit enzyme activity of pathogenic bacteria, affecting normal cell metabolism [62].

Although surfactin did not demonstrate significant antimicrobial activity against planktonic *P. fluorescens* cells (Figure 3A), effectively inhibited biofilm formation from this strain at all the tested concentrations (Figure 3B). Shen et al. [63] reported that, compared with rhamnolipid biosurfactants, SDS had a stronger effect on planktonic cells; here, the same occurred with surfactin, and the biofilms of all the tested strains were inhibited when both products were added directly to the culture media, indicating their great potential to protect surfaces from microbial biofilms.

5. Conclusions

In conclusion, the present study demonstrates the great potential of surfactin produced from a strain of *B. velezensis* H2O-1 as an antimicrobial, antiadhesion, antibiofilm, and anticorrosion agent. Compared with commercial products, bioproducts can replace synthetic products for specific purposes to avoid food contamination by food pathogens and spoilage microorganisms. Additionally, bioproducts are a more sustainable alternative.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/pr13010018/s1. Figure S1: Scanning electron microscopy images of mature biofilms of *L. monocytogenes* ATCC 19112 (left), *L. monocytogenes* ATCC 7644 (center), and *P. fluorescens* ATCC 13525 (right) on polystyrene (top) and stainless steel AISI 304 (bottom); Figure S2: Reproduction of the Table 1 from our previous work (page 175); Figure S3: Stainless steel 304 AISI control (top-left), unconditioned (top-right), conditioned with surfactin (bottom-left), and conditioned with SDS (bottom-right); Figure S4: Stainless steel 430 AISI control (top-left), unconditioned (top-right), conditioned with surfactin (bottom-left), and conditioned with SDS (bottom-right); Figure S5: Carbon steel control (top-left), unconditioned (top-right), conditioned with surfactin (bottom-left), and conditioned with SDS (bottom-right); Figure S6: Galvanized steel control (top-left), unconditioned (top-right), conditioned with surfactin (bottom-left), and conditioned with SDS (bottom-right).

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